

Jaypee University of Engineering and Technology, Guna
B.Tech. CSE, Semester V
Advanced Programming Lab - II (CS215/ 18B17CI573)
Lab Exercise 6

1. Design a Java program to simulate a **Smart Parking Management System** using a package named **parking**. The package should contain three classes: **Vehicle**, **ParkingSlot**, and **ParkingTransaction**. The **Vehicle** class should have private data members **vehicleNumber** and **vehicleType**, a constructor to initialize these values, and a method to display vehicle details. The **ParkingSlot** class should have private data members **slotId** and **isOccupied**, a constructor to initialize them, and a method to display slot details. The **ParkingTransaction** class should have private data members **transactionId**, **vehicle** (an object of **Vehicle**), and **parkingSlot** (an object of **ParkingSlot**), with a constructor to initialize them and a method to display transaction details. In the main class, which should be outside the package, create multiple objects of **Vehicle**, **ParkingSlot**, and **ParkingTransaction**, display the details of all vehicles, parking slots, and transactions, and implement logic to ensure that a vehicle is assigned only to a free parking slot while marking the slot as occupied.

2. Design a Java program to create a package **package1** containing a class **Employee** with a data member **salary**, a constructor to initialize it, and a method to display the salary. In the same package, define a subclass **Manager** that inherits **Employee**, uses a constructor to initialize the salary, and provides a method to display the manager's salary. In another package **package2**, create a subclass **Intern** that also inherits **Employee**, initializes the salary through its constructor, and displays the intern's salary. Finally, write a main class outside both packages to create objects of **Manager** and **Intern** and display their respective salaries.

3. Design a Java project to simulate a **Student Performance Tracker** system to explore the accessibility of **private**, **public**, and **default members** in different classes and packages. Create a package named **studentdata** containing a class **Student** with three data members: **studentId** (default) to store the student's ID, **studentName** (public) to store the student's name, and **marks** (private) to store the total marks. Include a constructor to initialize all members and a method to display student details. In the same package, create a class **Teacher** with a method to update the marks of a student object and display the student's ID and name, while attempting to access all members of the **Student** class to observe which are accessible. Next, create another package named **analytics** with a class

PerformanceAnalyzer that calculates the grade of a student based on the marks and attempts to access all members of the **Student** class to note accessibility across packages. In the main class, which should be outside both packages, create multiple **Student** objects, update marks using **Teacher** methods, analyze grades using **PerformanceAnalyzer**, and display all student details along with their grades. While performing these tasks, students should carefully observe and document the scope and accessibility of private, public, and default members, and understand how encapsulation and access specifiers affect data visibility across classes and packages.

4. Design a Java program to manage a small inventory system **using only arrays**. Create a package named **inventory** and define a class **Product** with the following private data members: **productId** (int), **productName** (String), **quantityAvailable** (int), and **unitPrice** (double). Provide public getter and setter methods for all data members. Implement methods to update the stock when new items are added or sold, apply a discount to the product price, and display the details of a product. In the main class, create multiple **Product** objects, and implement a simple menu-driven system with the following tasks:
 - a. Add a new product to the inventory.
 - b. Update the stock of an existing product.
 - c. Apply a discount to a product.
 - d. Display the details of all products.

Ensure the program continues running until the user chooses to exit. While writing the program, consider making it user-friendly and realistic for a small inventory scenario.

5. Solve the following questions using any language:
 - a. <https://www.geeksforgeeks.org/problems/count-the-numbers2359/1?page=1&category=Arrays&company=Morgan%20Stanley&sortBy=submissions>
[Asked in Morgan Stanley]
 - b. <https://www.geeksforgeeks.org/problems/maximum-index-1587115620/1?page=3&category=Arrays&sortBy=submissions>
[Asked in Amazon, VMWare, Microsoft and MakeMyTrip]